27.7.26

EXP 2 : STUDY OF NETWORK CABLES

AIM:

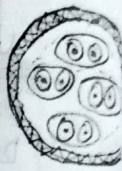
To study the different types of network cables.

a. **Understand different types of network cable**

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Sheilded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use	Image
UTP	Category 3	10 bps	Advantages: • cheaper in cost. • Easy to install as they have a smaller overall diameter. Disadvantages: • More prone to (EMI) electro-magnetic interface and noise.	10 Base - T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	

STP	Category 6, 6a	10 Gbps	Advantages: • Sheilded • Faster than UTP • Less susceptible to noise and interference. Disadvantages: • Expensive • Greater installation + effort.	Gigabit Ethernet, 10G Ethernet widely used in centers.	
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100m)	 
Coaxial cable.	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile. Disadvantages: • Limited distance • Cost • Size is bulky.	Speed of signal is 500m Television network High speed internet connections	
Fibre optics cable	Single mode Multi mode	100 Gbps	• High speed • High bandwidth • High security • Long distance Disadvantages: • Expensive • Required skillers.	Maximum distance of fibre optics cable is around 100 meters.	

RESULT:

Thus, the study of different types of cable is executed successfully.